MIL-N-8922/1A
25 July 1988
Superseding
MIL-N-8922/1 (AS)
23 August 1982

MILITARY SPECIFICATION SHEET

NUT, SELF-LOCKING, ASSEMBLED WASHER, ALLOY STEEL, 220 KSI Ftu, 450°F, (12-SPLINE WRENCHING ELEMENT)

This specification is approved for use by all Agencies of the Department of Defense.

The complete requirements for acquiring the nuts described herein shall consist of this document and the latest issue of MIL-N-8922.

1. Application criteria: The nuts shall be used in accordance with the limitations of MS33588.

2. Tooling: During installation, wrenches in accordance with MIL-W-8982 shall be used for hand or power driven nuts.

3. Military installation: Navy - NAVAIR-01-1A-8
Air Force - T.O. 1-1A-8

4. General requirements:

4.1 Material: Nut material shall be H-11 alloy steel in accordance with AMS 6485 or AMS 6487. Washer material shall be carbon steel in accordance with AMS 5085.

4.2 Heat treatment: The nut shall be heat treated to Rockwell C 40-45 in accordance with MIL-H-6875. The washer hardness shall be Rockwell C 40-45 KNOOP (402-466).

4.3 Surface texture shall be 125 in accordance with ANSI B46.1.

4.4 Coating: The nut and washer shall be cadmium plated in accordance with QQ-P-416, Type II, Class 2.

4.5 Lubrication: The nut shall be lubricated with lubricant in accordance with MIL-L-8937. Cetyl alcohol lubricants, shall be soluble in the cleaner specified in MIL-L-87132.

4.6 Threads shall be in accordance with MIL-S-8879 before lubrication.

4.7 Wrenching element shall be in accordance with MS33787.

FSC 5310

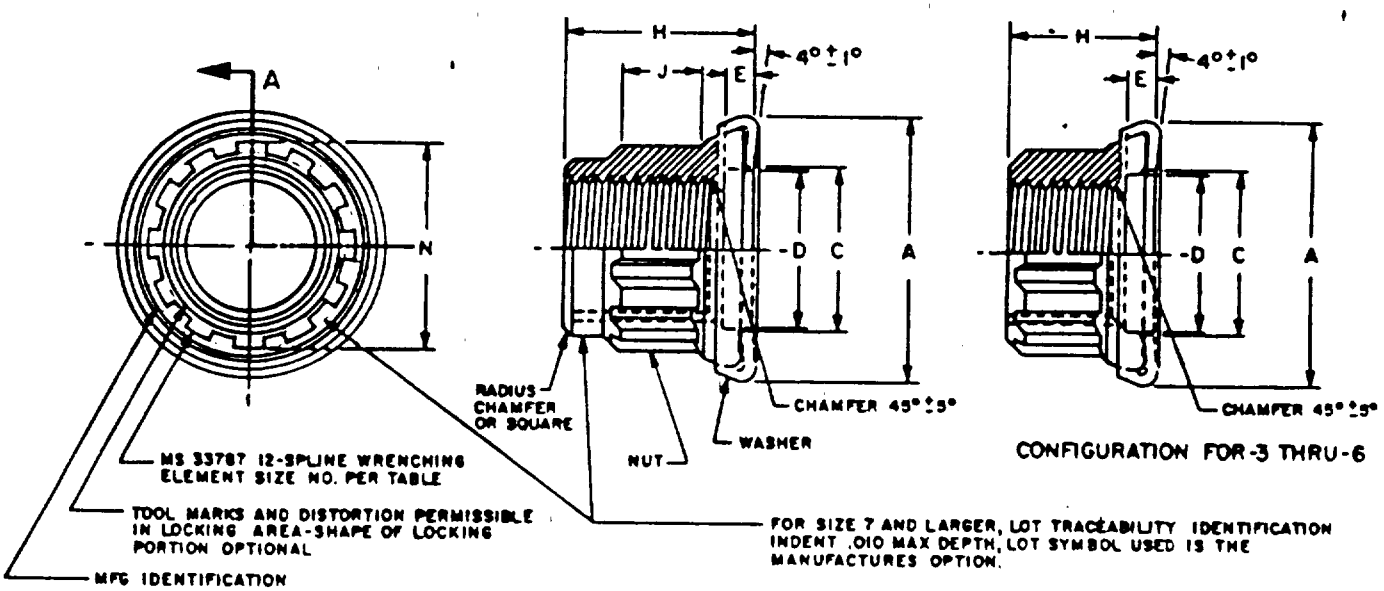
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4.8 Geometric tolerance: Perpendicularity - the bearing surface shall be normal with pitch diameter of thread within Y values T.I.R. when checked in accordance with MIL-N-8922.

4.9 Concentricity: N diameter shall be concentric to pitch diameter of thread within X values T.I.R.

4.10 Dimensions shall be in inches. Dimensions apply before lubrication.

5. Illustration:



SECTION - A

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6. Table I. Dimensions.

Size Dash No.	Thread Size MIL-S-8879	MS33787 (12 Spline) Element Number	A Max	C ±.005	D ±.005	E +.010 -.005
3	.1900-32UNJF-3B	8	.460	.288	.282	.077
4	.2500-28UNJF-3B	10	.560	.353	.347	.077
5	.3125-24UNJF-3B	12	.660	.418	.412	.077
6	.3750-24UNJF-3B	14	.760	.485	.479	.077
7	.4375-20UNJF-3B	18	.860	.552	.546	.077
8	.5000-20UNJF-3B	20	1.000	.636	.621	.077
9	.5625-18UNJF-3B	22	1.100	.765	.750	.087
10	.6250-18UNJF-3B	24	1.310	.840	.825	.087
12	.7500-16UNJF-3B	30	1.480	.987	.972	.092
14	.8750-14UNJF-3B	34	1.690	1.111	1.096	.097
16	1.0000-12UNJF-3B	38	1.834	1.253	1.238	.097
18	1.1250-12UNJF-3B	44	2.100	1.425	1.410	.097
20	1.2500-12UNJF-3B	48	2.250	1.500	1.485	.097
22	1.3750-12UNJF-3B	52	2.480	1.650	1.635	.097
24	1.5000-12UNJF-3B	56	2.690	1.780	1.765	.097

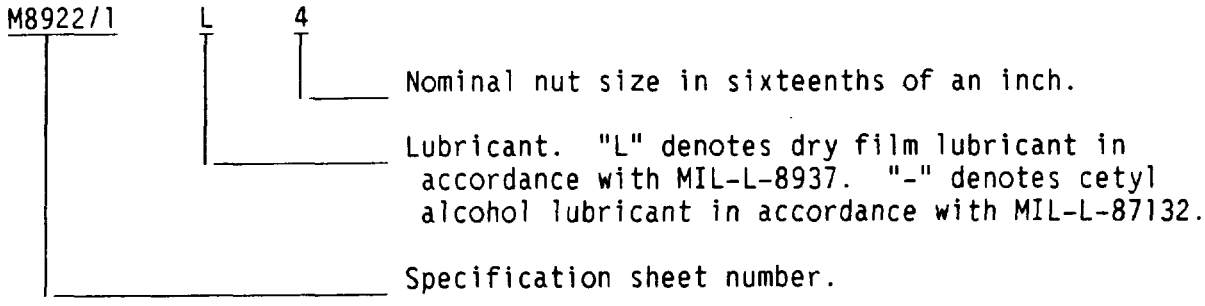
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Size	H		J	N	X	Y	Ultimate Axial Tensile Strength Lb Min (a)
	Max	Min	Min	Max			
3	.280	.260	.080	.308	.003	.004	4,970
4	.332	.312	.100	.381	.003	.004	8,890
5	.400	.380	.120	.456	.003	.005	14,100
6	.478	.458	.152	.531	.003	.006	20,900
7	.525	.505	.168	.679	.003	.008	28,300
8	.603	.583	.190	.753	.003	.008	37,800
9	.672	.652	.230	.828	.004	.009	47,900
10	.734	.714	.250	.901	.004	.010	59,900
12	.879	.859	.310	1.124	.004	.012	86,900
14	1.012	.992	.350	1.270	.005	.014	119,000
16	1.140	1.120	.400	1.419	.006	.016	155,000
18	1.337	1.317	.450	1.641	.007	.019	198,000
20	1.524	1.504	.500	1.788	.007	.021	247,000
22	1.659	1.639	.550	1.937	.008	.023	301,000
24	1.847	1.827	.600	2.085	.008	.023	361,000

(a) Axial tensile strength determined from formula Wa = FtuA. Where: A is the cross sectional area in square inches, based on the maximum pitch diameter of bolt thread, Ftu is 220 KSI and Wa is the axial tensile strength in pounds.

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7. Part number example and code:



8. Administrative and contractual provisions:

8.1 For design feature purposes this document takes precedence over procurement document referenced herein.

8.2 Referenced documents shall be of the issue in effect on date of invitation for bid.

Custodians:
Army - AV
Navy - AS
Air Force - 99

Preparing activity:
Navy - AS
(Project No. 5310-1257)

Review Activities:
Air Force - 82
DLA - 11

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